Group Simulation Project: Linear Search Algorithm

Course: Data Structures & Algorithms

Repository Name: `DSA-Roleplay-Ch2-6`

GitHub Repository: **https://github.com/pro-cluster11/DSA-Roleplay-Ch2-6**

Video Link: **https://t.me/+EvOfiz5Rg0szNmM0**

1. Topic Rationale & Team Roles

Chosen Topic: Linear Search (Chapter 2)

Rationale: Linear Search was selected because it provides a clear, step-by-step sequential pattern that translates effectively into a physical simulation. It allows us to demonstrate best-case $O(1)$ and worst-case $O(N)$ execution bounds using simple node comparisons and pointer shifts.

Member Role	Assigned Responsibility
Camera / Narrator / Scribe	Operates camera, explains algorithmic logic, and logs operations live.
Pointer / Search Index	Holds index arrow and steps between array nodes sequentially.
Data Node 0	Holds card value **4** (Index 0).
Data Node 1	Holds card value **7** (Index 1).
Data Node 2	Holds card value **11** (Index 2).
Data Node 3	Holds card value **15** (Index 3).

2. Step-by-Step Operation Log

Initial Array State: `[4, 7, 11, 15]`

Operation 1: Best-Case Search (Target = 4)

Step	Current Index	Value Checked	Comparison	Result	Cumulative Comparisons
1	0	4	`4 == 4`	**Match Found**	1

Operation 2: Average-Case Search (Target = 11)

Step	Current Index	Value Checked	Comparison	Result	Cumulative Comparisons
1	0	4	`4 == 11`	No match, advance pointer	1
2	1	7	`7 == 11`	No match, advance pointer	2
3	2	11	`11 == 11`	**Match Found**	3

Operation 3: Worst-Case Search - Element Present (Target = 15)

Step	Current Index	Value Checked	Comparison	Result	Cumulative Comparisons
1	0	4	`4 == 15`	No match, advance pointer	1
2	1	7	`7 == 15`	No match, advance pointer	2
3	2	11	`11 == 15`	No match, advance pointer	3
4	3	15	`15 == 15`	**Match Found**	4

Operation 4: Worst-Case Search - Element Absent (Target = 20)

Step	Current Index	Value Checked	Comparison	Result	Cumulative Comparisons
1	0	4	`4 == 20`	No match, advance pointer	1
2	1	7	`7 == 20`	No match, advance pointer	2
3	2	11	`11 == 20`	No match, advance pointer	3
4	3	15	`15 == 20`	No match, advance pointer	4
5	Out of Bounds	—	Loop Ends	**Target Not Found**	4

3. Summary Tally of Operations

Scenario	Target	Array Size (N)	Total Comparisons	Time Complexity
Best Case	4	4	1	$O(1)$
Average Case	11	4	3	$O(N)$
Worst Case (Present)	15	4	4	$O(N)$
Worst Case (Absent)	20	4	4	$O(N)$
Total Operations Executed			**12 Comparisons**	—

4. Critical Reflection on Role-Play Limitations

1. **Pacing and Live Audio Synchronization:** Real-time human interaction created slight delays between the pointer's physical step, the narrator's vocalization, and the scribe logging tally marks, unlike a computer processor's synchronous clock speed.
2. **Visual Framing & Legibility:** Maintaining all four static nodes, the dynamic pointer, and the live tally board within a single wide camera shot without sacrificing text legibility required specific physical positioning compromises.
3. **Simulating Out-of-Bounds Checks:** Abstract concepts like checking `index < array.length` required physical boundary markers and dummy signs to signal array termination to the viewer.
4. **Distinguishing Index vs. Value:** Human participants occasionally risked confusing their ordinal physical position (Index 0, 1, 2, 3) with the numerical data value printed on their cards during rapid comparisons.